Progressive Autonomy as Preference Learning

A Formalization of Trust Calibration for Agentic Tool Use

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Abstract

We formalize trust calibration for agentic tool use (deciding when an automated agent's proposed action may execute autonomously versus require human approval) as a preference-learning problem. A policy gateway maintains a Gaussian-process posterior over a latent human risk-tolerance function, observed through a probit likelihood on binary approve/deny feedback, and escalates to the human exactly where the approval outcome is most uncertain. We show this is structurally an instance of Preferential Bayesian Optimization, inheriting its inference machinery (approximate Gaussian-process classification) and its sample-efficiency argument (uncertainty-targeted querying), while differing in objective: classifying an action space into allow/block/ask regions rather than optimizing a design.

1 Background and Related Work

Deciding how much autonomy to delegate to an automated system is a classical human-in-the-loop control problem; the technical core here, recovering a human's latent acceptability function from sparse binary feedback, is *preference learning*. Chu and Ghahramani [6] introduced Gaussian-process preference learning, placing a Gaussian-process (GP) prior over a latent utility and linking observed human choices to it through a probit likelihood. We adopt exactly this construction, specialized to unary approve/deny feedback. The same latent-utility-with-probit model underlies preference-driven sequential decision making: González et al. [8] formalized *Preferential Bayesian Optimization* (PBO), embedding GP preference learning in the query loop of Bayesian optimization [3, 16]. Our policy gateway is structurally an instance of this framework, differing only in objective (classifying an action space rather than optimizing a design), as Table 1 makes precise.

The inferential machinery is classical GP classification: a GP prior, a non-Gaussian probit likelihood, and an analytically intractable posterior approximated by the Laplace method or Expectation Propagation [10], developed comprehensively by Rasmussen and Williams [13]. Treating the human query budget as a scarce resource makes the ASK region an *acquisition* rule in the sense of active learning [15] and Bayesian optimization [16]: interruptions are spent where the expected information about the allow/block boundary is greatest. Finally, drifting risk tolerance is a non-stationarity problem; we model it with a time-decaying kernel component in the spirit of non-stationary covariance functions [12] and, for abrupt shifts, Bayesian online changepoint detection [1].

This progressive view of autonomy has deep roots in the trust-in-automation literature: Lee and See [9] characterized appropriate reliance as the alignment of trust with actual system trustworthiness, and de Visser et al. [7] extended this to *longitudinal* trust calibration in human–robot teams, precisely the dynamic our time-decaying kernel (§6) is meant to capture. The concern has resurfaced sharply for large-language-model agents, where graduated autonomy has emerged as an explicit deployment axis alongside raw capability [11]. Recent work catalogs the risks of increasingly agentic systems [4], argues for visibility and oversight mechanisms over deployed agents [5], and proposes governance practices in which a human retains approval authority over consequential actions [17]. These accounts are largely qualitative and taxonomic: they argue *why* graduated autonomy matters and *what* should be governed, but leave the escalation policy itself as a fixed, hand-specified tier. Our contribution is the missing mechanism: a learning rule that lets the auto-approve/escalate boundary adapt from human feedback rather than being set by hand.

2 Setup

At each decision point $t = 1, 2, \ldots$, the policy gateway observes a proposed agent action $a_t \in \mathcal{A}$ and an execution context $c_t \in \mathcal{C}$, where:

$$a_t = (\texttt{tool\_name},\ \texttt{args},\ \texttt{target\_resource}), \tag{1}$$

$$c_t = (\texttt{repo\_state},\ \texttt{task\_desc},\ \texttt{session\_history}). \tag{2}$$

A human supervisor provides binary feedback $y_t \in \{0, 1\}$ (deny/approve). We write $x_t := (a_t, c_t) \in \mathcal{X} = \mathcal{A} \times \mathcal{C}$ for the joint input.

3 Latent Risk Tolerance

Definition 1 (Risk tolerance function)**.** *There exists a latent function $f\colon \mathcal{X} \to \mathbb{R}$ encoding the human's risk tolerance, such that approval probability follows a probit observation model:*

$$\Pr(y = 1 \mid x) = \Phi\big(f(x)\big), \tag{3}$$

where Φ is the standard normal CDF.

Remark 1. *This is structurally identical to the observation model in Preferential Bayesian Optimization [8] and, more fundamentally, in Gaussian-process preference learning [6]. In standard PBO, the human expresses a preference between two candidates (x_i, x_j) via $\Pr(x_i \succ x_j) = \Phi\big(f(x_i) - f(x_j)\big)$. Here, the comparison is against an implicit internal threshold: the human approves when $f(x)$ exceeds their acceptable-risk boundary. The unary case is a degenerate pairwise comparison against a fixed reference point.*

4 Gaussian Process Prior and Posterior

Place a GP prior over f [13]:

$$f \sim \mathcal{GP}\big(\mu_0,\ k(x, x')\big). \tag{4}$$

The kernel k decomposes over the input structure. A natural choice is a product kernel:

$$k(x, x') = k_{\text{tool}}(a, a') \cdot k_{\text{ctx}}(c, c') \cdot k_{\text{time}}(t, t'), \tag{5}$$

where:

- k_{tool} encodes similarity between actions (e.g., shared tool name, overlapping argument patterns, same reversibility class),
- k_{ctx} captures context similarity (same repository, file type, task category),
- k_{time} handles non-stationarity (see §6).

After observing $\mathcal{D}_N = \{(x_t, y_t)\}_{t=1}^N$, the posterior is:

$$p(f \mid \mathcal{D}_N) \ \propto \ \mathcal{GP}(\mu_0, k) \ \cdot \prod_{t=1}^{N} \Phi\big(f(x_t)\big)^{y_t} \big(1 - \Phi\big(f(x_t)\big)\big)^{1-y_t}. \tag{6}$$

This is analytically intractable due to the non-Gaussian likelihood. Approximate inference proceeds via the Laplace approximation [13] or Expectation Propagation [10], the same machinery used in PBO.

Remark 2 (Practical realization)**.** *For the unary model, the Laplace approximation of Rasmussen and Williams [13] (Algorithm 3.1 there) reduces inference to a short Newton iteration to the posterior mode, after which the predictive approval probability admits the closed form* $\mathbb{E}\big[\Phi(f(x_*))\big] \approx \Phi\big(\mu_*/\sqrt{1+\sigma_*^2}\big)$*, with* μ_* *and* σ_*^2 *the latent posterior mean and variance at* x_**. At the scale of decision points in a single project (*N *on the order of* 10^2 *to* 10^3*), this exact computation is inexpensive and needs no sparse approximation, which keeps the implementation a transparent instance of the cited machinery rather than a wrapper around a general optimization stack. A maintained off-the-shelf alternative is the* `PairwiseGP` *model of BoTorch [2], a deployable PBO implementation; being pairwise, it recovers the unary setting only through the degenerate comparison against a fixed reference of Remark 1, at the cost of a heavier dependency. This choice affects only the posterior: the gateway rule, the non-stationary drift component, and any baselines are unchanged.*

5 The Policy Gateway Decision Rule

Given the posterior predictive distribution at a new point x_*:

$$\hat{p}(x_*) := \mathbb{E}_{f|\mathcal{D}_N}\big[\Phi(f(x_*))\big], \tag{7}$$

the gateway applies a three-tier decision:

$$\text{decision}(x_*) = \begin{cases} \text{ALLOW} & \text{if } \hat{p}(x_*) > \tau_{\text{high}}, \\ \text{BLOCK} & \text{if } \hat{p}(x_*) < \tau_{\text{low}}, \\ \text{ASK} & \text{otherwise.} \end{cases} \tag{8}$$

The ASK region $[\tau_{\text{low}}, \tau_{\text{high}}]$ plays the role of an *acquisition function* [15, 16]: the system queries the human precisely where the model is most uncertain about the approval outcome, maximizing the expected value of information per human interruption.

Remark 3. *The target operating point (85–90% auto-approve, 10–15% human escalation) emerges naturally as the posterior concentrates. Early in a project, most actions fall in the* ASK *region. As the model learns, the* ASK *band narrows, and the auto-approve rate rises without any manual threshold tuning.*

6 Non-Stationarity

Human risk tolerance drifts: early in a project the supervisor is cautious; as trust accumulates for familiar patterns, they become permissive. Model this via a time-decayed kernel component:

$$k_{\text{time}}(t, t') = \exp\Big(-\frac{|t - t'|}{\lambda}\Big), \tag{9}$$

where $\lambda > 0$ is a lengthscale controlling the forgetting rate. This gives recent approve/deny signals more weight, a principled analog of the intuition that "agents earn trust over time."

For computational efficiency, an equivalent effect can be achieved through a sliding window of the most recent W observations, or via online changepoint detection when the supervisor's behavior shifts abruptly (e.g., moving to a new codebase).

7 Correlated Generalization

A key advantage over a naïve contextual bandit (which treats each (a, c) independently) is *correlated generalization* through the kernel. Concretely:

- Approving `write_file` to `/workspace/src/` transfers evidence to `write_file` to `/workspace/test/`, since k_{tool} and k_{ctx} assign high similarity.
- Denying `execute_sql` with a `DROP` argument propagates caution to `execute_sql` with `TRUNCATE`, without the human having to deny each variant individually.
- A new tool with no interaction history inherits the prior μ_0, which maps to ASK, the fail-safe default.

8 Connection to PBO

The mapping between the trust calibration problem and Preferential Bayesian Optimization [8] is summarized in Table 1.

Component	**PBO (Optimization)**	**Trust Calibration (Policy)**
Input space $\mathcal{X}$	Design parameters	(action, context) pairs
Latent function f	Objective to maximize	Risk tolerance to learn
Human feedback	Pairwise preference $x_i \succ x_j$	Unary approve/deny
Observation model	$\Phi(f(x_i) - f(x_j))$	$\Phi(f(x))$
Acquisition	Next query to evaluate	Next action to escalate
Goal	Find $x^* = \arg\max f$	Learn the ALLOW/BLOCK boundary

Table 1: Structural correspondence between PBO and trust calibration.

The "preferential" aspect is literal: the human is expressing preferences over what the agent should be allowed to do. The same mathematical machinery (GP priors, probit likelihoods, approximate posterior inference) transfers directly. The difference is the objective: PBO seeks to *optimize*, while trust calibration seeks to *classify* the action space into allow/block/ask regions with minimal human queries.

9 Datasets and Evaluation

Empirically grounding the gateway requires data pairing proposed agent actions with human approve/deny judgments. The closest public resource is R-Judge [18], which supplies multi-turn agent interaction records annotated by humans with binary safe/unsafe labels across a range of risk scenarios; it is a natural source for a cold-start prior μ_0 and for calibrating k_{tool} and k_{ctx}. Broader agent-safety benchmarks such as Agent-SafetyBench [19] and ToolEmu [14] widen the action and context coverage, but their risk labels are produced by automated judges rather than per-action human approval, so they are better suited to stress-testing the learned boundary than to fitting f itself.

A structural gap remains: no public dataset captures the *longitudinal, per-supervisor* signal that the non-stationarity model (§6) assumes. Existing corpora provide single-shot, aggregated annotations and do not track an individual supervisor's risk tolerance drifting over the course of a project. Validating the time-decaying kernel therefore requires either a controlled user study or a simulation with deliberately drifting synthetic annotators, with R-Judge serving as the static initializer. We treat this as an inherent limitation of currently available data rather than a deficiency of the model.

We accordingly exercise the formulation in a controlled simulation with a known ground-truth oracle that instantiates Definition 1 and the drift of §6, the standard evaluation protocol for Preferential Bayesian Optimization. Reproducible code, figures, and a generated report are provided under `experiment/`. Fitting the Laplace GP-probit gateway online and scoring it prequentially confirms the central behaviors: the ASK band narrows as the posterior concentrates, the auto-approve rate rises toward the target operating point at a multiplicative reduction in human interruptions relative to always escalating, correlated generalization transfers evidence to unqueried action–context combinations, and the posterior tracks both gradual drift and an abrupt changepoint. The simulation also surfaces one honest caveat about §5: the rule "query inside the ASK band," taken literally, is sample-efficient only while the target is stationary. Under an abrupt changepoint it has an exploration deficit, since regions the posterior has grown confident about leave the ASK band and are never re-queried; the time-decayed kernel down-weights stale evidence but does not by itself generate new probes. A recency-aware or information-theoretic acquisition criterion (for example an expected-information or BALD rule with a forgetting term) is the natural remedy and is left to future work.

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